

Computational Acceleration of Elastohydrodynamic Lubrication Solvers via Progressive Mesh Densification

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Introduction

Background and Literature Review

- Elastohydrodynamic Lubrication (EHL) Solvers are commonly used for **modelling complex interactions between lubricated surfaces** such as roller bearings (Ardah et al., 2023). Although these solvers provide highly detailed solutions at small errors, **they demand large computation times**
- Researchers have actively developed several techniques to optimise these solvers over many decades. More recently, a novel approach known as Progressive Mesh Densification (PMD) (Pu, Wang & Zhu, 2015) has been deployed which **passes the EHL solution through a series of meshes, each mesh finer than the previous**.
- This study will assess the effectiveness of **implementing PMD in MATLAB** through the use of the internal interpolation function as shown in **Figure 1**.

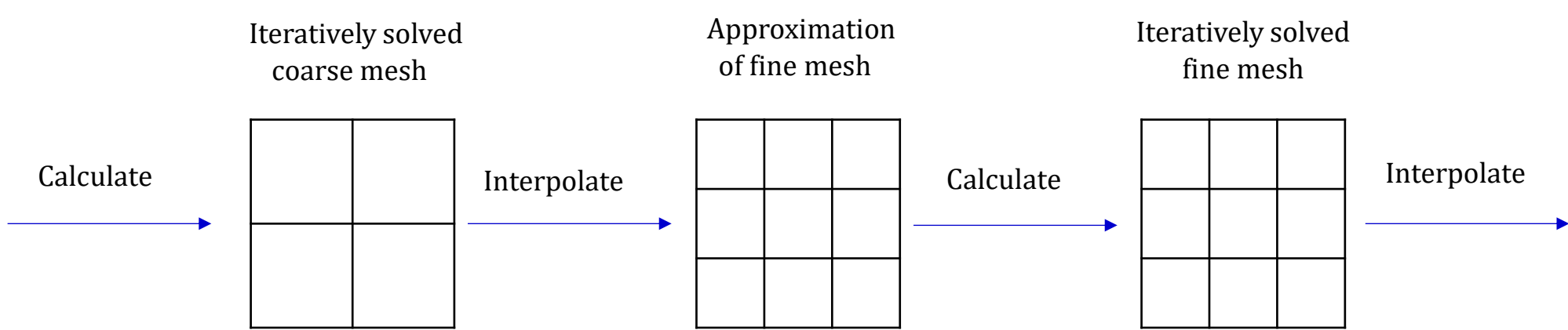


Figure 1. visually demonstrates the cycle implemented enabling progressive mesh densification (PMD) in MATLAB

Results and Discussion

The EHL solver was adapted to export the calculated values in the form of CSV files that could be used in data analysis. As seen in **Figure 2**, the mesh size has a significant impact on the accuracy of the resulting graphs.

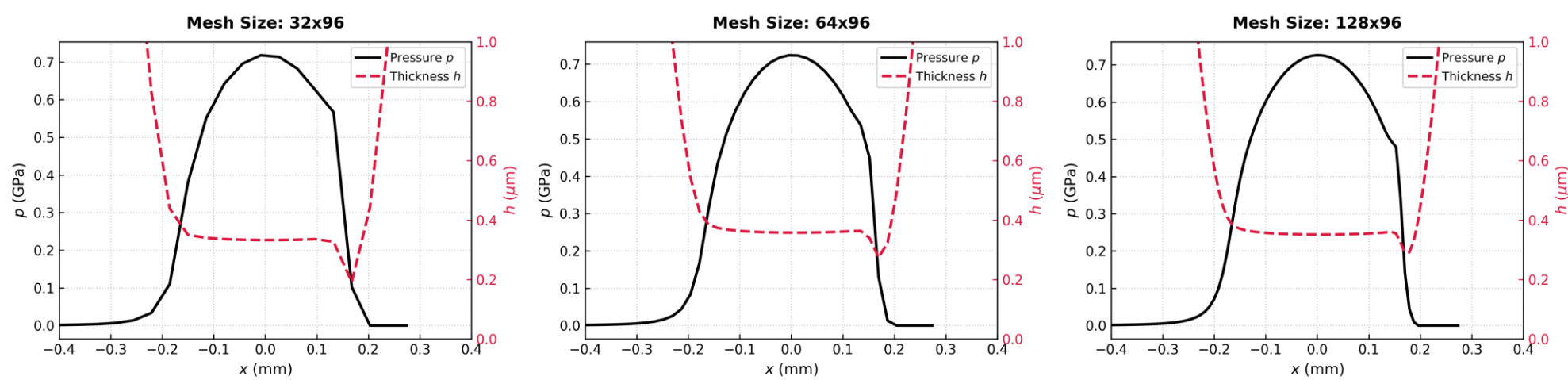


Figure 2. Three mesh sizes used to demonstrate the impact of mesh size on the accuracy of solver results.

It is evident that the larger mesh size is required to provide improved results for accurate lubricant modelling. However larger mesh sizes will drastically impact the computation time as commonly discussed in literature. The implementation of PMD for this project will make use of the 2D interpolation function available in MATLAB as shown in **Figure 3**.

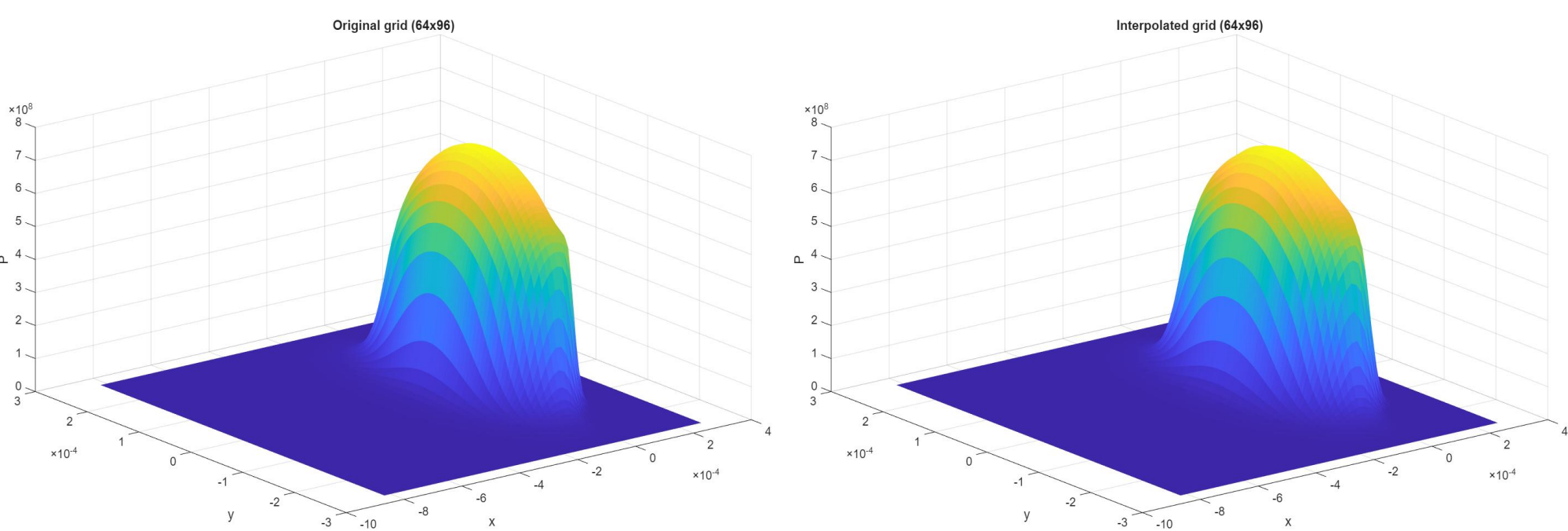


Figure 3. Demonstration of the 2D interpolation function used on a mesh of 32x96 to interpolate a mesh of 64x96 as shown on the right-hand side. For comparison, a standard calculated 64x96 mesh can be found on the left-hand side of the figure.

Conclusions and Future Work

The implementation of PMD into the EHL solver proved to be incredibly successful by providing significant reductions in the computation time required to achieve an accurate set of solutions. It is evident that the implementation of PMD is best suited for very large mesh sizes, demonstrating most drastic reductions in time for large meshes.

Future work will be focused on collecting further data for a more reliable time prediction model. Additionally, roughness and temperature gradient can be introduced for more complex modelling.

Methodology

A complete numerical approach was adopted using MATLAB simulations. The base model has been previously developed to solve for a singular defined mesh that iteratively computes a solution closer to the real solution. The code is defined to stop once the iterative solutions are within a defined error of the previous.

A set of base results were collected by executing the original code with a range of defined meshes. **Collected data contained hydrodynamic pressure, film thickness and the execution time**

The progressive mesh densification framework was implemented by utilising the **interpolation functionality in MATLAB**. This function can interpolate a coarse mesh to a finer mesh with a marginal computation time.

The interpolated mesh provides a much significantly improved approximation for the finer mesh, enabling the **defined error to be reached** at significantly fewer computational iterations.

The 2D interpolation function had been placed to execute prior to the iterative solver to greatly reduce the time taken to reach the maximum allowed error. A plot comparing the computation time for the standard calculation against the implemented PMD method can be found in **Figure 4**.

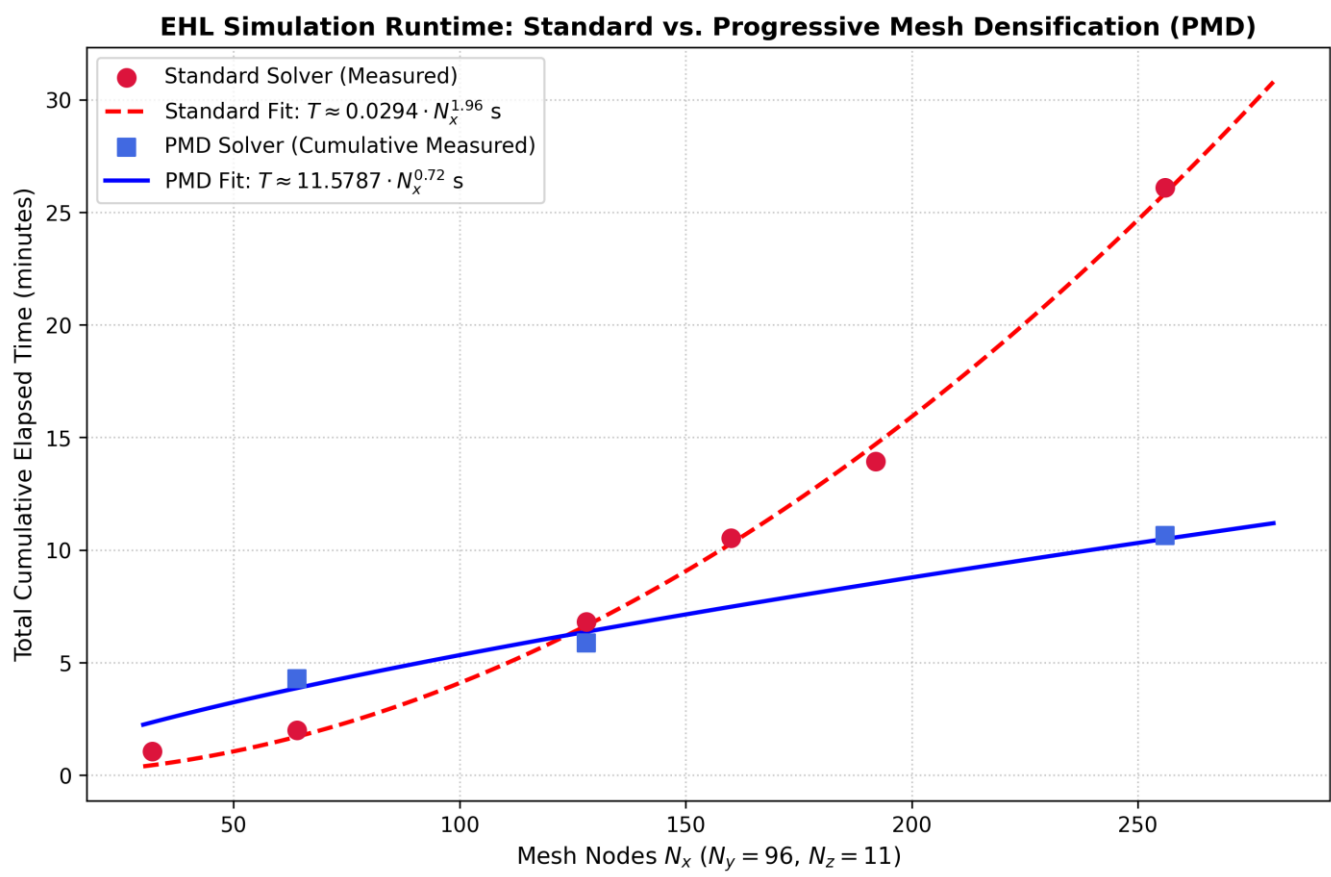


Figure 4. Plot of recorded times at different mesh sizes. This includes both the standard solver times along with the solver times for the implemented PMD.

The results demonstrated in **Figure 4** were used to extract a numerical model fit capable of estimating the time saving impact of introducing PMD. These were then used to fill **Table 1**.

Table 1. contains a set of estimated times calculated in minutes for the execution of a series of meshes.

Mesh size	Standard (min)	PMD (min)	% reduction
256x96	25.7	10.5	59.1
512x96	100.1	17.2	82.8
1024x96	389.4	28.4	92.7

References

- Ardah, S., Profito, F.J., Reddyhoff, T. & Dini, D. (2023) Advanced modelling of lubricated interfaces in general curvilinear grids. *Tribology International*. 188, 108727. doi:[10.1016/j.triboint.2023.108727](https://doi.org/10.1016/j.triboint.2023.108727).
- Pu, W., Wang, J. & Zhu, D. (2015) Progressive Mesh Densification Method for Numerical Solution of Mixed Elastohydrodynamic Lubrication. *Journal of Tribology*. 138 (021502). doi:[10.1115/1.4031495](https://doi.org/10.1115/1.4031495).